Sample Lecture 2: Conditional Probability

A synthetic lecture handout for testing the learn plugin. Not from any course.

2.1 Definition

For events A and B with $P(B) > 0$, the conditional probability of A given B is

$$P(A \mid B) = P(A \cap B) / P(B).$$

2.2 Multiplication rule

For any events A and B with $P(B) > 0$: $P(A \cap B) = P(A \mid B) \cdot P(B)$.

2.3 Law of total probability

If $B_1, \ldots, B_n$ partition Ω and each $P(B_i) > 0$, then $P(A) = \Sigma_i P(A \mid B_i) P(B_i)$.

2.4 Bayes' theorem

$P(B \mid A) = P(A \mid B) P(B) / P(A)$, for $P(A) > 0$ and $P(B) > 0$.

Handwritten note (sample)

P(A | B) is not P(B | A) in general.
Example: P(positive | ill) = 0.99
but P(ill | positive) can be small
when the illness is rare.